

Optimization Techniques in Power System (BT EE610)



हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

Fuzzy Logic Algorithm

By
Dr. Manish Kumar
Asth. Professor
Department of Electrical Engineering
SOE&T, CUH

What is Fuzzy Logic?

- Fuzzy Logic (FL) is a method of reasoning that resembles human reasoning. The approach of FL imitates the way of decision making in humans that involves all intermediate possibilities between digital values YES and NO.
- The conventional logic block that a computer can understand takes precise input and produces a definite output as TRUE or FALSE, which is equivalent to human's YES or NO.
- Fuzzy logic is a form of many-valued logic; it deals with reasoning that is approximate rather than fixed and exact. In contrast with traditional logic theory, where binary sets have two-valued logic: true or false, fuzzy logic variables may have a truth value that ranges in degree between 0 and 1
- Fuzzy logic has been extended to handle the concept of partial truth, where the truth value may range between completely true and completely false. Furthermore, when linguistic variables are used, these degrees may be managed by specific functions.

Fuzzy sets

Fuzzy sets - proposed by Lotfi Zadeh in 1965

- Set membership is graduated
- Degrees of belonging measured as real numbers between zero and one
- Boundaries of the set are soft, not crisp

Definition

A fuzzy set A over the universe for discourse X is a set of ordered pairs mapping domain elements their respective degrees of belonging measured as a real number between zero and one:

$$A = \{(x_1, 0.4), (x_2, 0.3), (x_3, 1), (x_4, 0.6)\}$$

Or using Zadeh's notation:

$$A = \{0.4/x_1 + 0.3/x_2 + 1/x_3 + 0.6/x_4\}$$

A fuzzy set A is usually expressed in terms of its membership function μ_A which maps domain elements (x) their respective degrees of belonging in the interval $[0, 1]$:

$$A = \{(x, \mu_A(x)) | x \in X\}$$

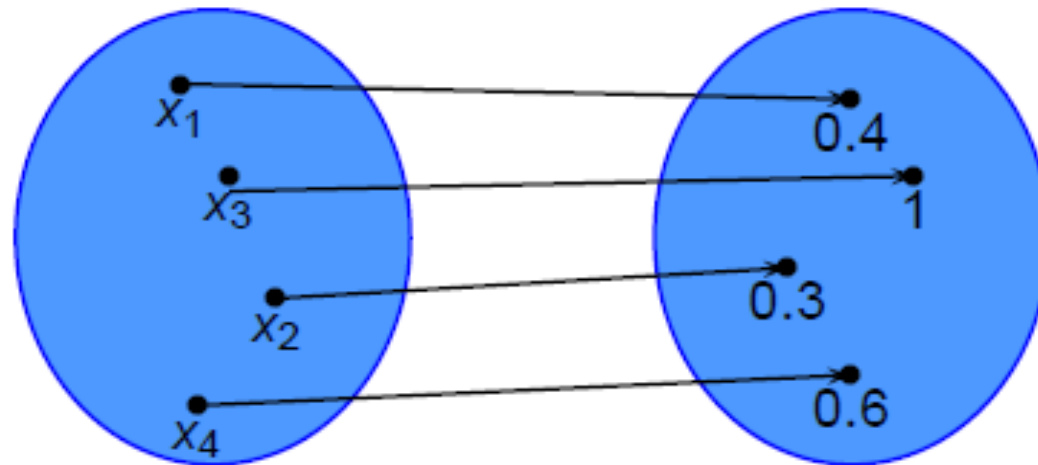


Lotfi Zadeh

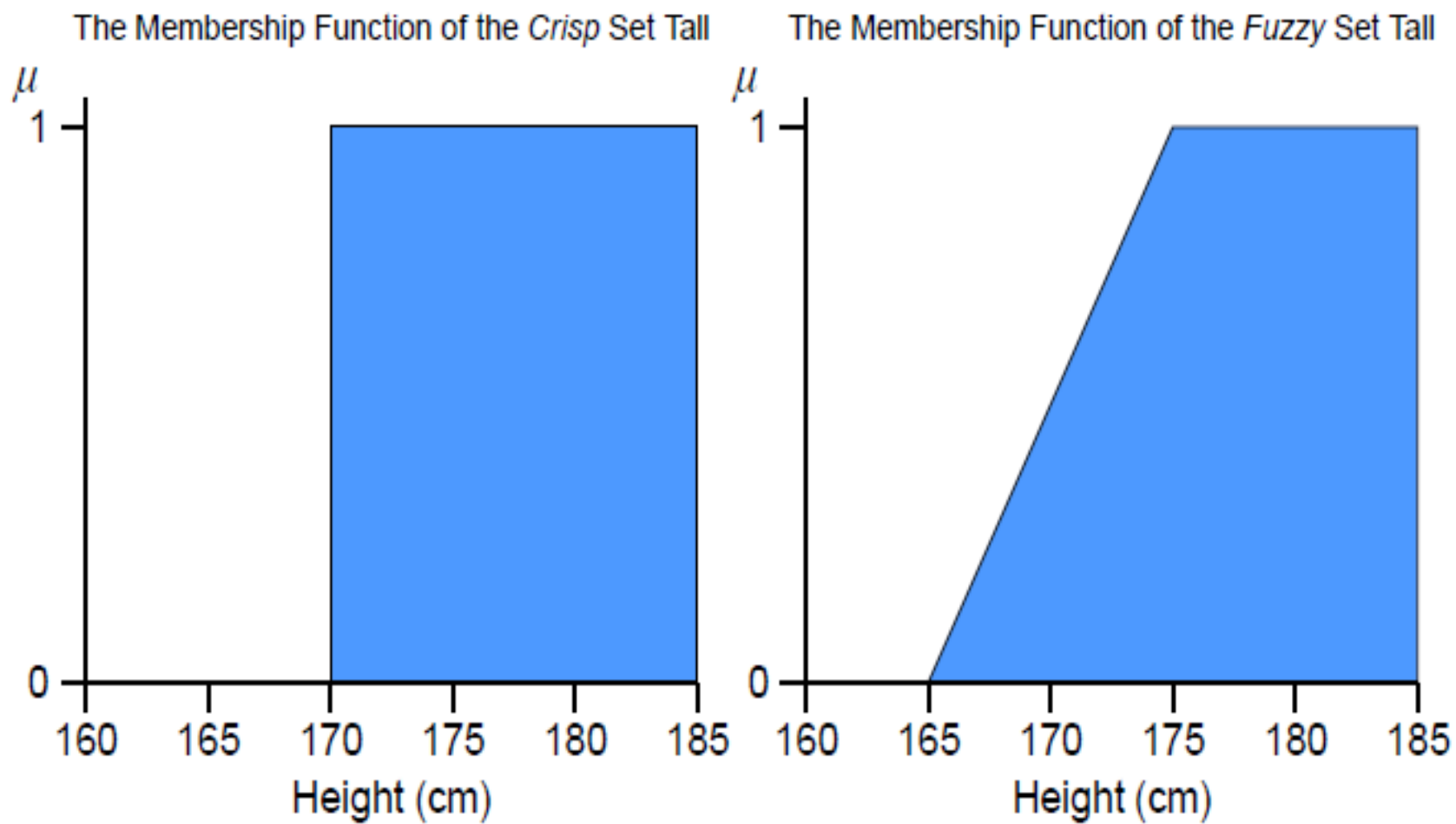
A Graphical Comparison with Crisp Sets

So a fuzzy set is a relation from some domain to real numbers:

$$A : X \times \{0, 1\}$$



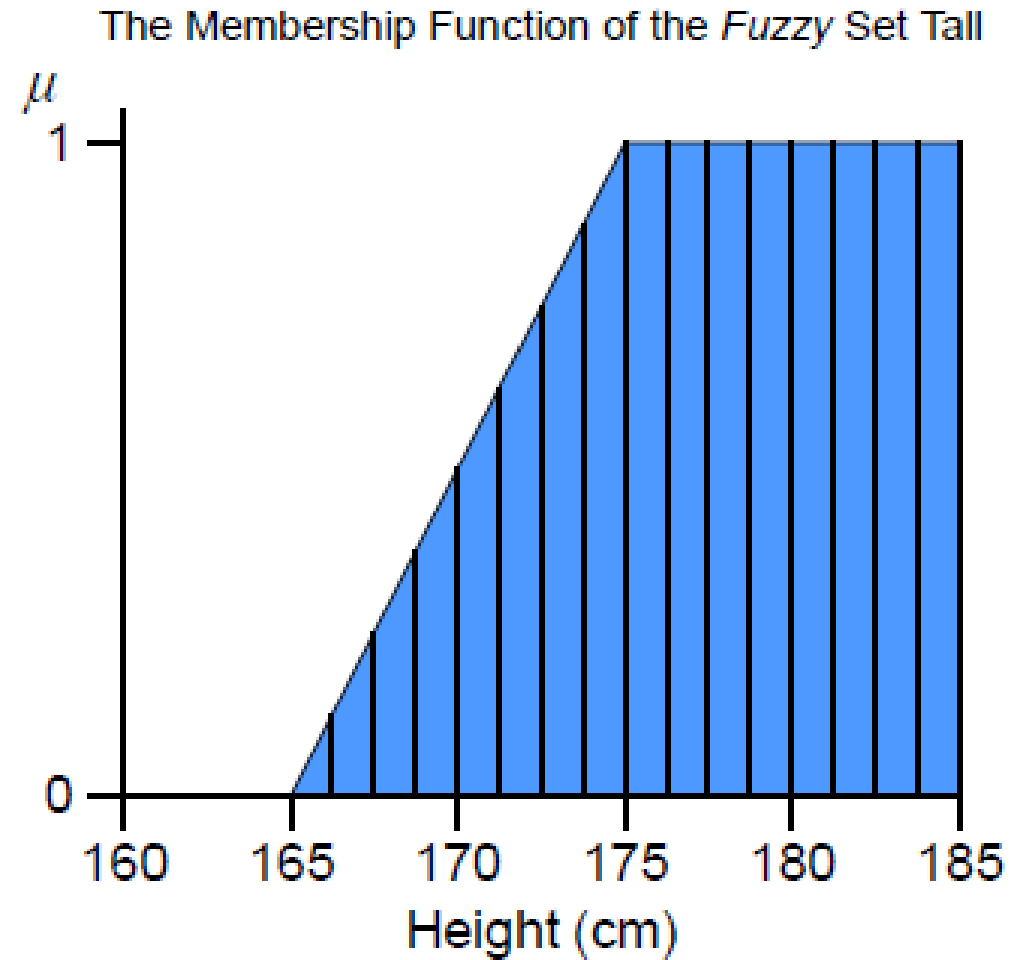
A Graphical Comparison with Crisp Sets



Implementation Reality

Computers don't like continuous functions

- Instead use discrete approximations
 - A number of ordered pairs mapping x values to the μ
- values to the μ*



Fuzzy Sets and Probability

A Cautionary Tale

- Quite different meanings
- Example - bottles of liquid:

Fuzzy Bottle



0.7 Drinkable

Probabilistic Bottle



0.7 Drinkable

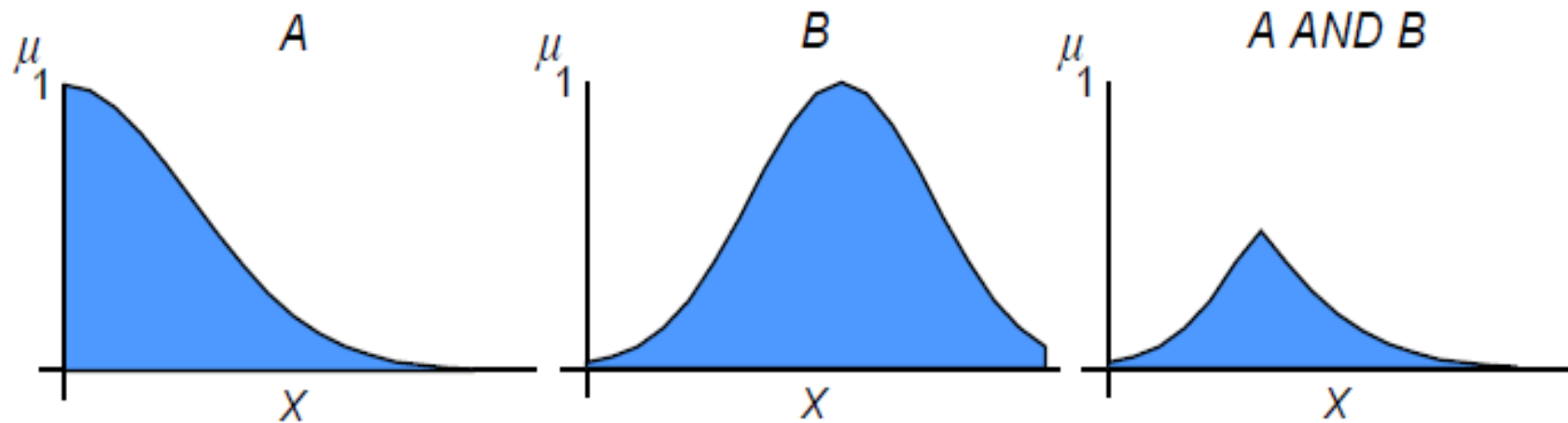
Fuzzy Logic Operators

- Logical operations of fuzzy sets are well defined
- Together these form fuzzy logic:
 - AND
 - OR
 - NOT
 - IMPLIES
- Crucial for rule based fuzzy logic systems

Logical AND

- Defined for each point in the membership function
- Extends Boolean AND
- Any t-norm but usually minimum:

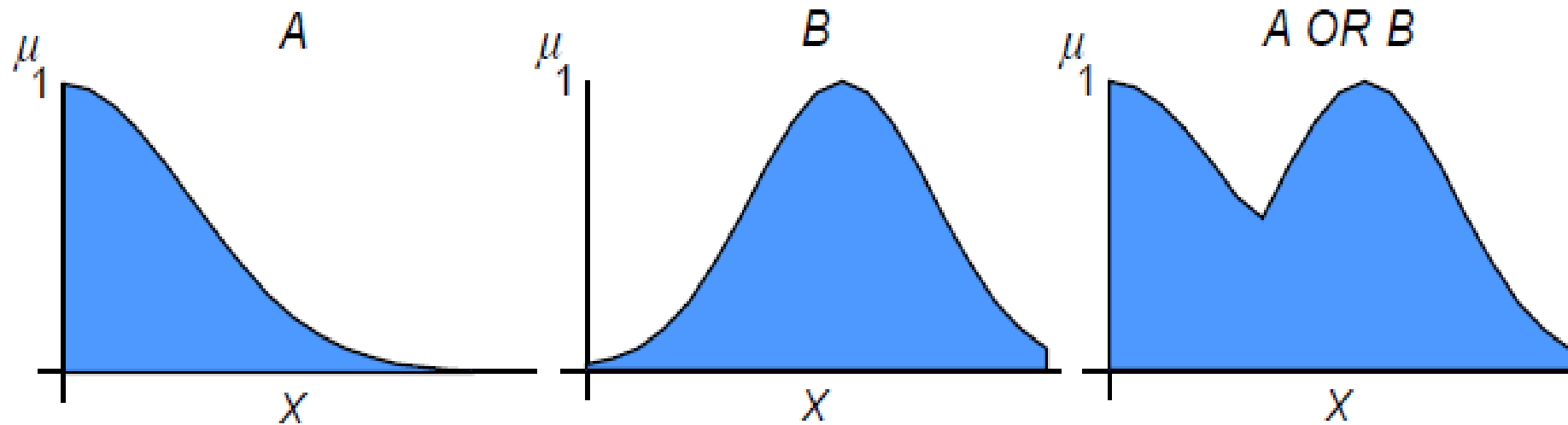
$$\mu_{A \text{ AND } B}(x) = \mu_A(x) \wedge \mu_B(x)$$



Logical OR

- Defined for each point in the membership function
- Extends Boolean OR
- Any t-norm but usually maximum:

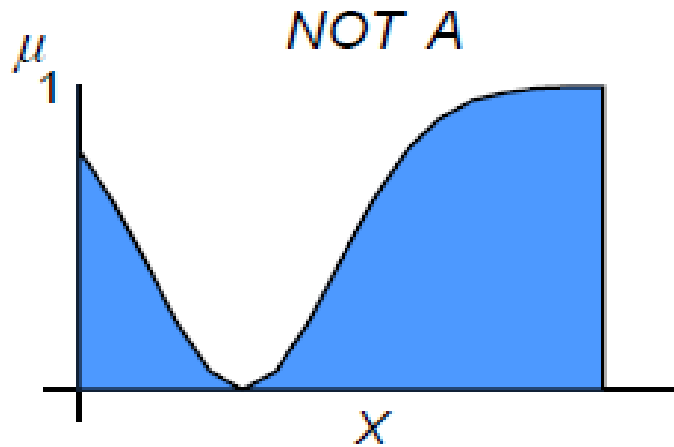
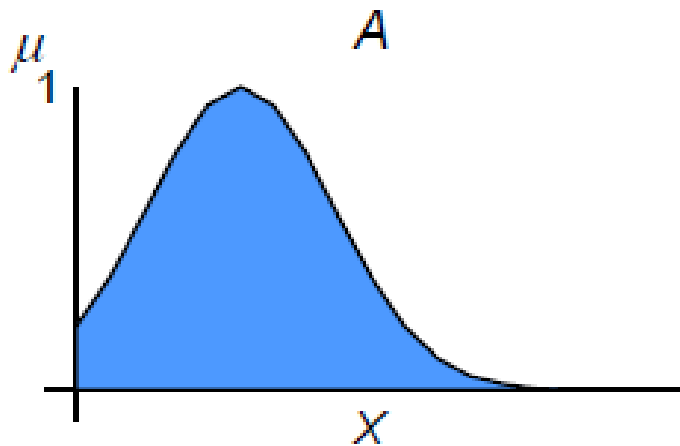
$$\mu_{A \text{ OR } B}(x) = \mu_A(x) \vee \mu_B(x)$$



Logical NOT

- Defined for each point in the membership function
- Extends Boolean NOT:

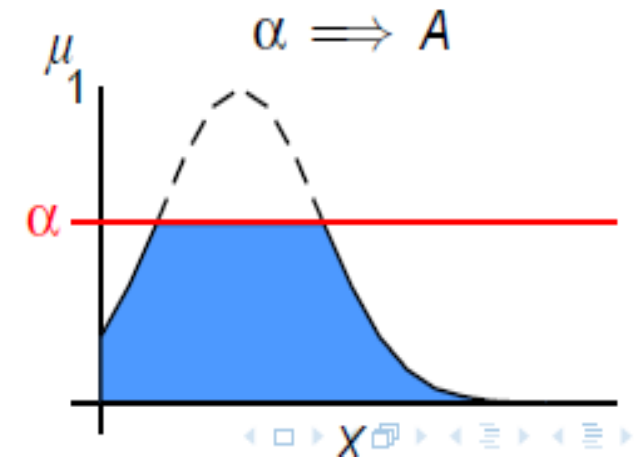
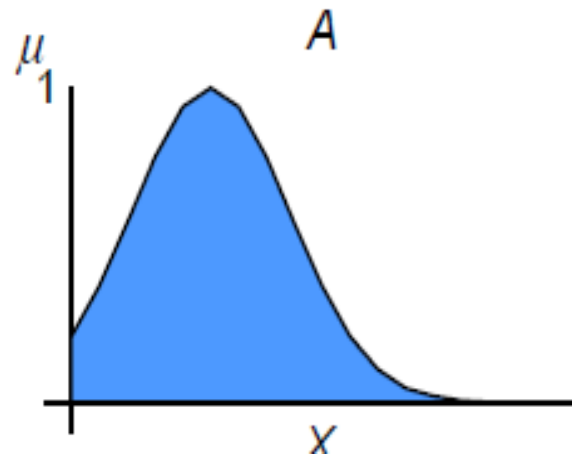
$$\neg \mu_A(x) = 1 - \mu_A(x)$$



Logical IMPLIES

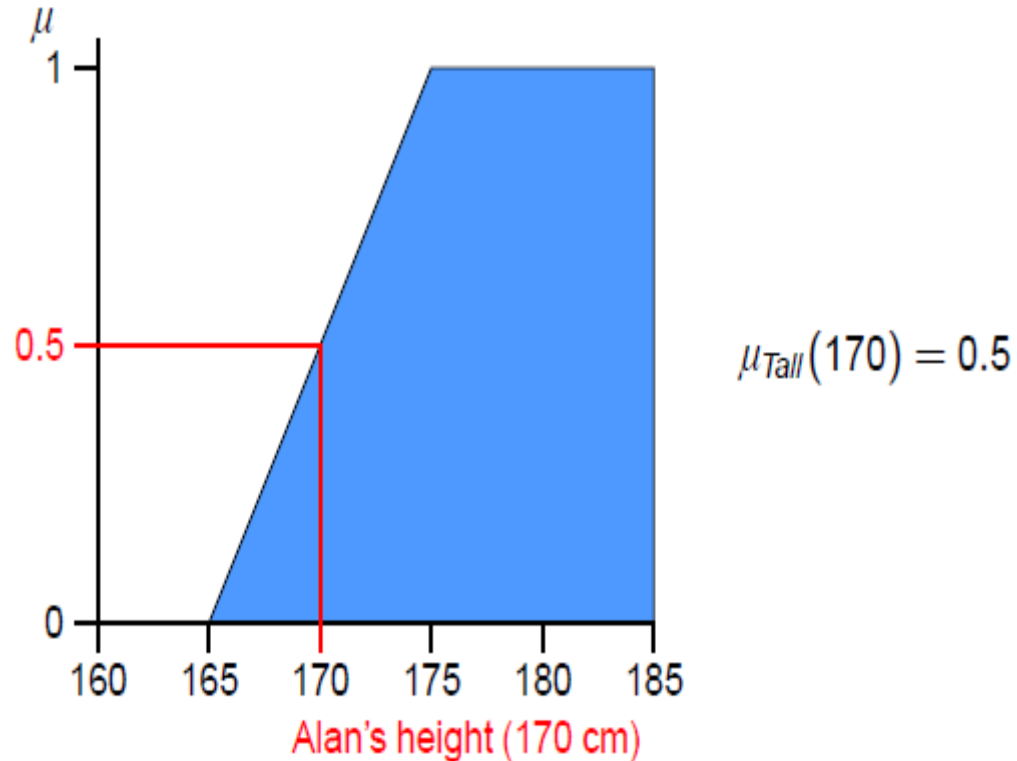
- Defined for each point in the membership function
- A variety of operators
- Most commonly used is generalized modus ponens:
 - Modus ponens: If X THEN Y . X , therefore Y
 - Generalized modus ponens: If X THEN Y . X to degree 0.6, therefore Y to degree 0.6
- Any t-norm but usually minimum:

$$\mu_{\alpha \Rightarrow A}(x) = \alpha \vee \mu_A(x)$$



Fuzzification

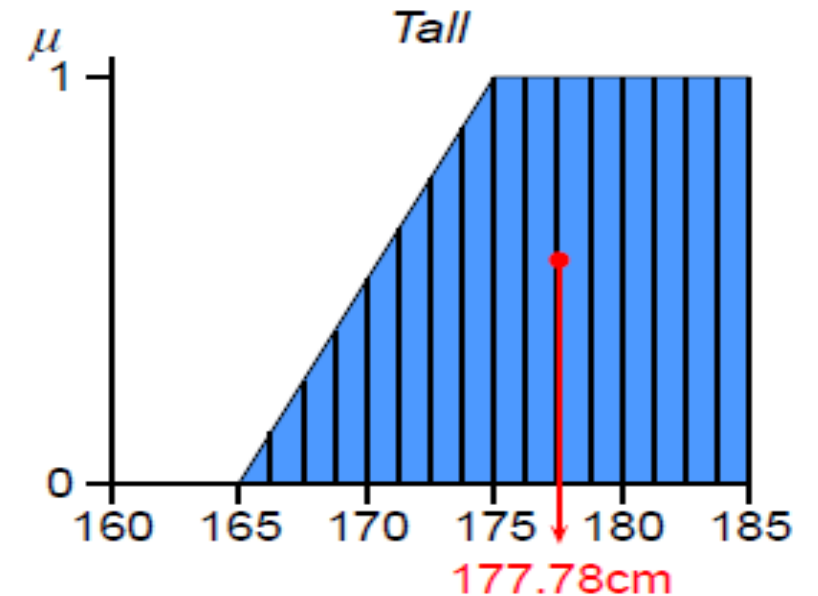
- The process of finding the membership grade of an input:



Defuzzification

- The process of reducing a fuzzy set to a single crisp value
- Centre of area is most commonly used:

$$C_A = \frac{\sum \mu_A(x)x}{\sum \mu_A(x)}$$
$$C_{Tall} = \frac{0.13 \times 166.25 + \dots + 1 \times 185}{0.13 + \dots + 1} = 177.78$$



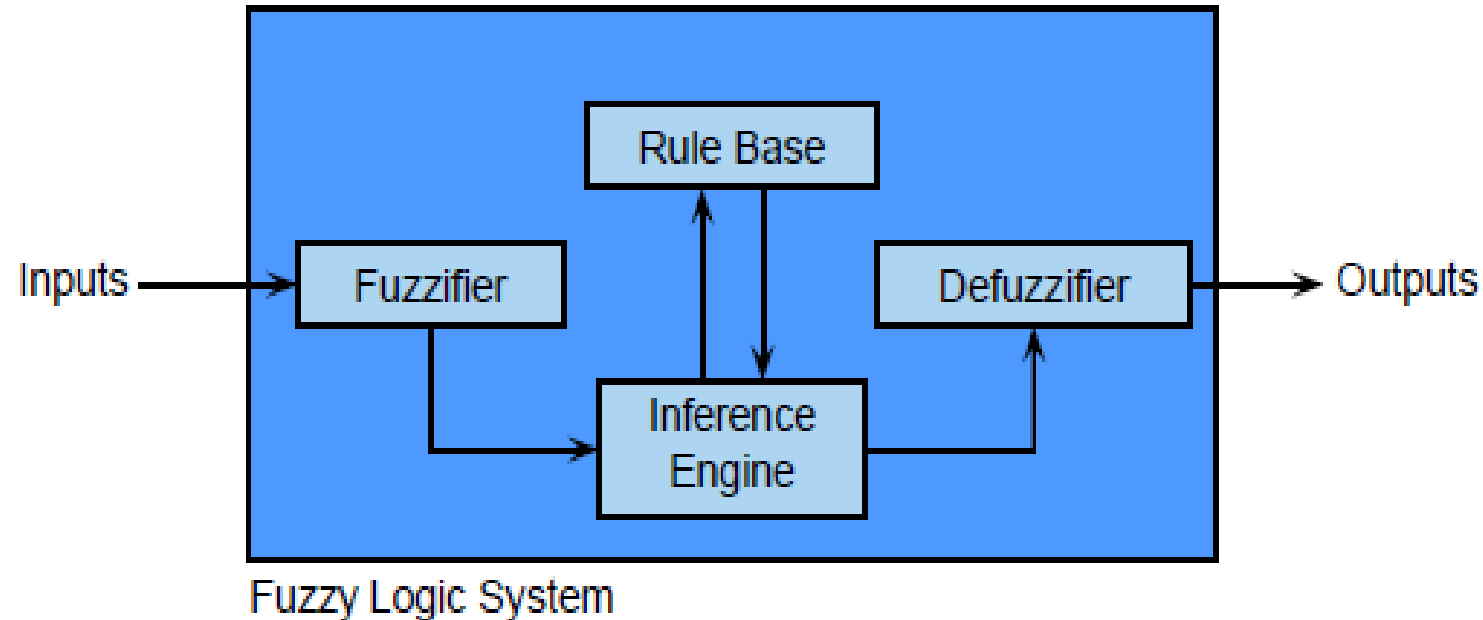
Fuzzy Logic Systems

Fitting it all Together

- Typically rule-based:

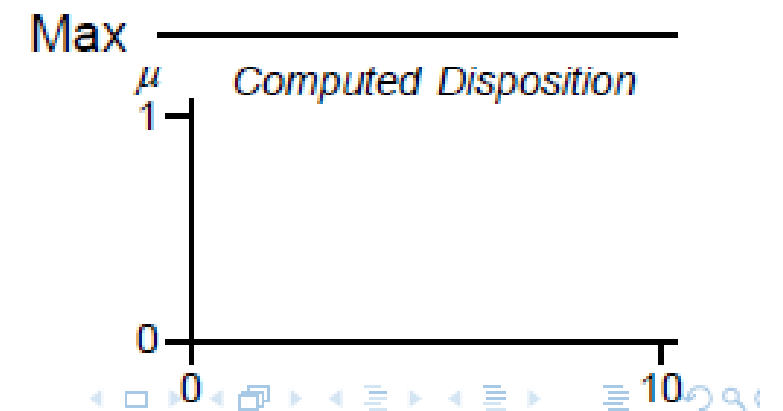
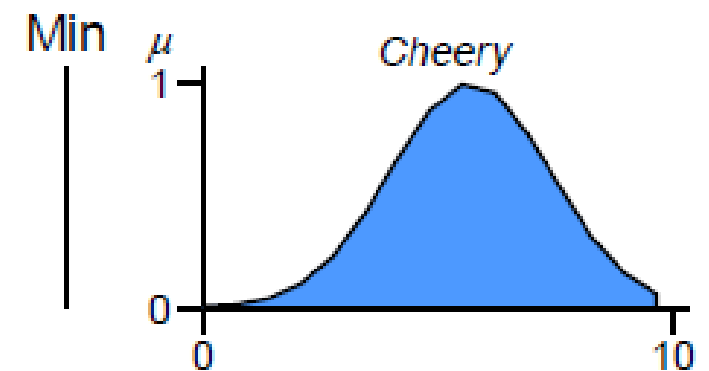
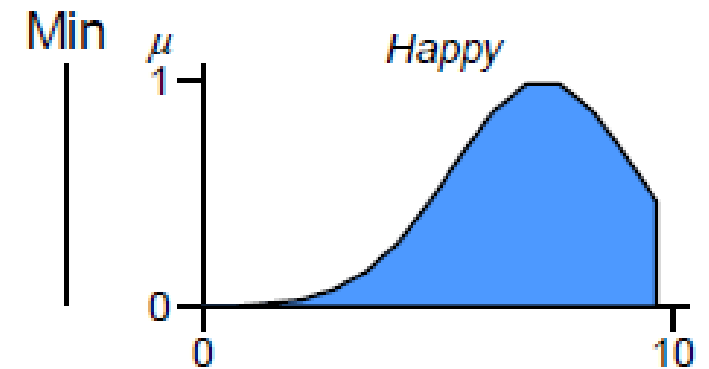
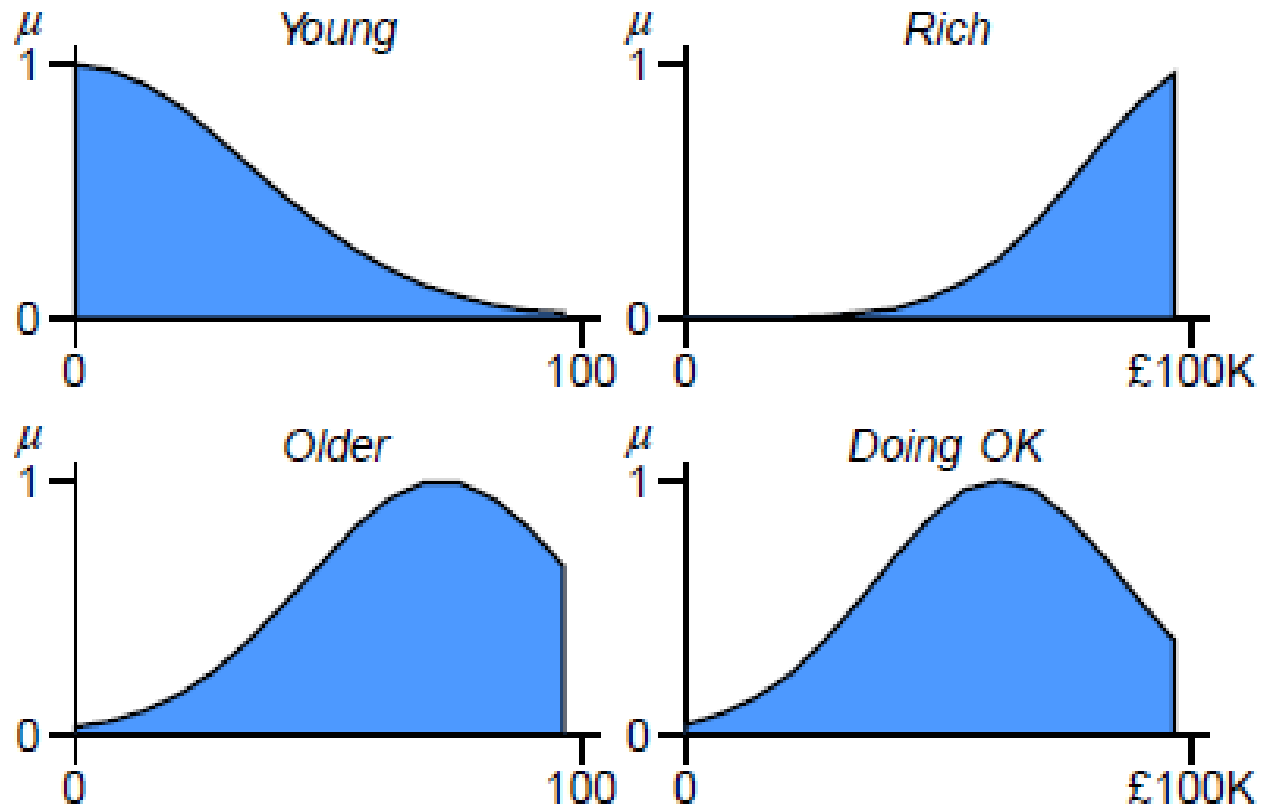
IF *age is Young* AND *wealth is Rich* THEN *disposition is Very Happy*

- Combine fuzzy sets with logical operators
- Crisp inputs, often crisp outputs:



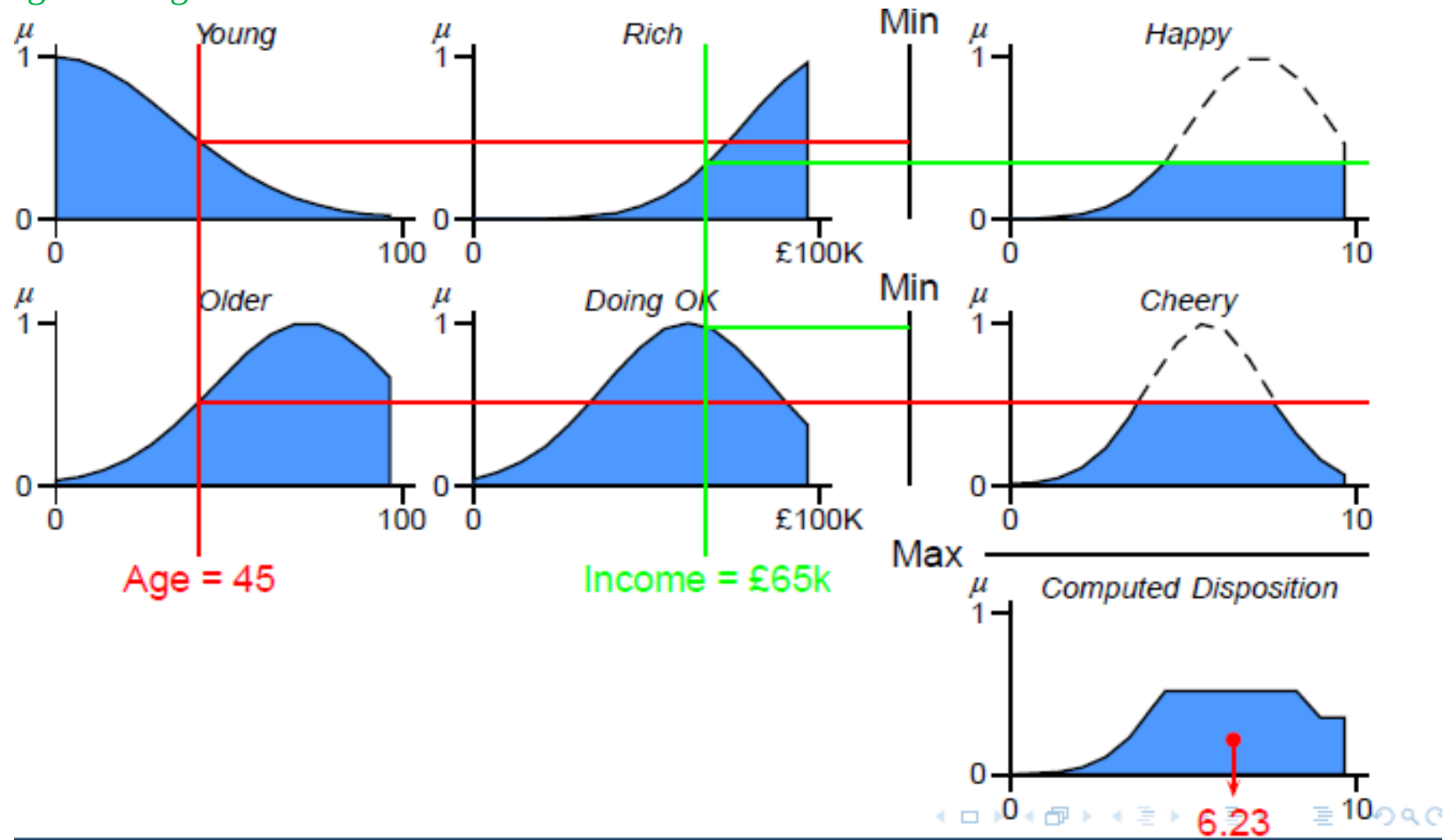
Fuzzy Logic Systems

Fitting it all Together



Fuzzy Logic Systems

Fitting it all Together



Fuzzy Logic Systems

What I haven't told you

Many other approaches:

- Logical operator choices
- Neuro-fuzzy systems
- Defuzzification operator choices
- Adaptive systems

Application Areas

Applied to a wide range of problems including:

- Industrial control
- Human decision making
- Image processing

Industrial Control

Control of marine diesel engines

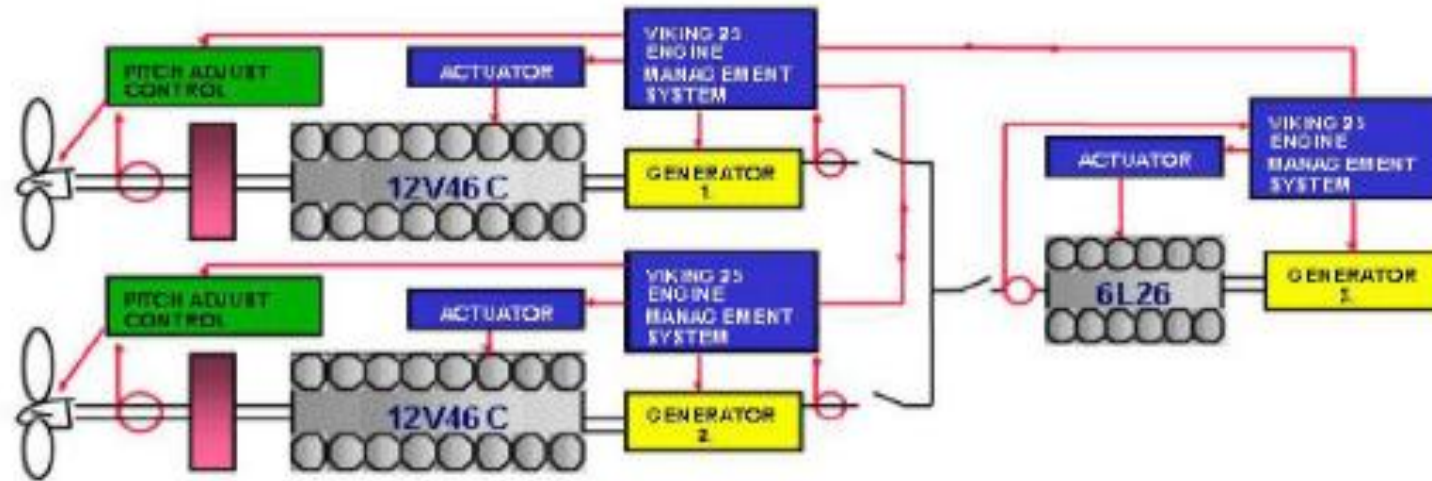
- MAN 9000kW Cathedral Engines
- Low overshoot tolerance
- Highly dynamic and uncertain environments
- Require robust and accurate control



Industrial Control

Control of marine diesel engines

- Typically three engines
- Two drive props and generators
- One solely for power generation

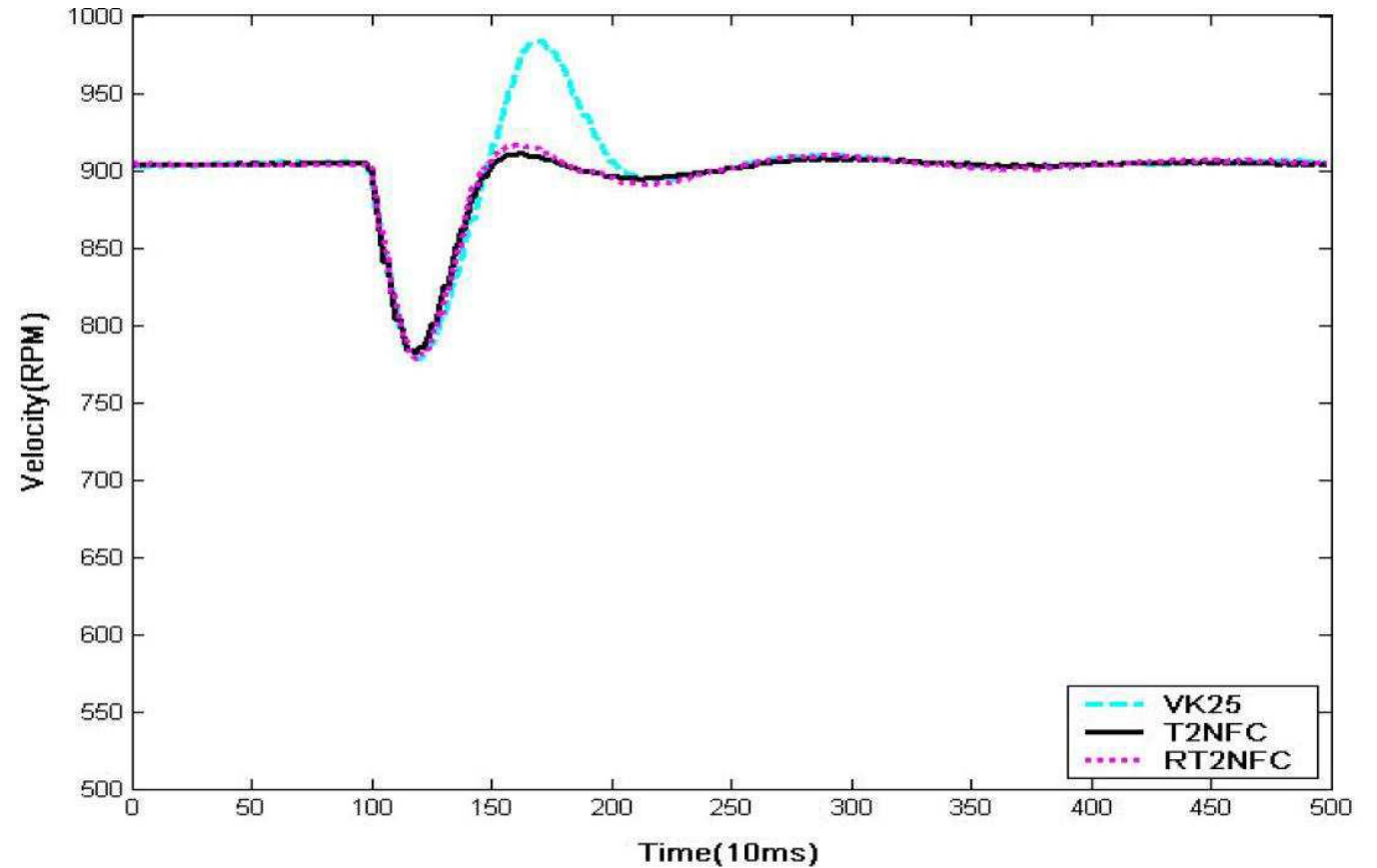


From Lynch, C. et al, Using Uncertainty Bounds in the Design of an Embedded Real-Time Type-2 Neuro-Fuzzy Speed Controller for Marine Diesel Engines, FUZZ-IEEE 2006.

Industrial Control

Control of marine diesel engines

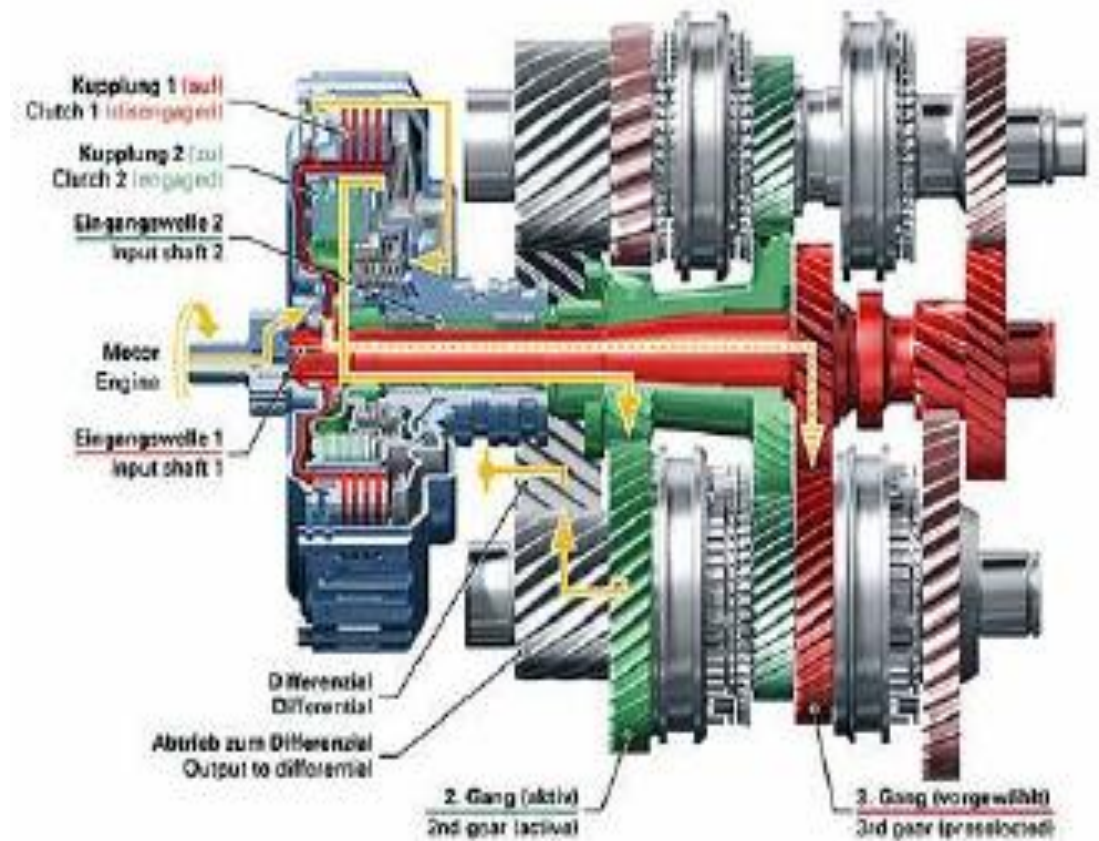
- VK25 is the current control system
- T2NFC and RT2NFC are both fuzzy
- Type-2 fuzzy sets
- Different defuzzification techniques



Human Decision Making

Volkswagen Direct-Shift Gearbox

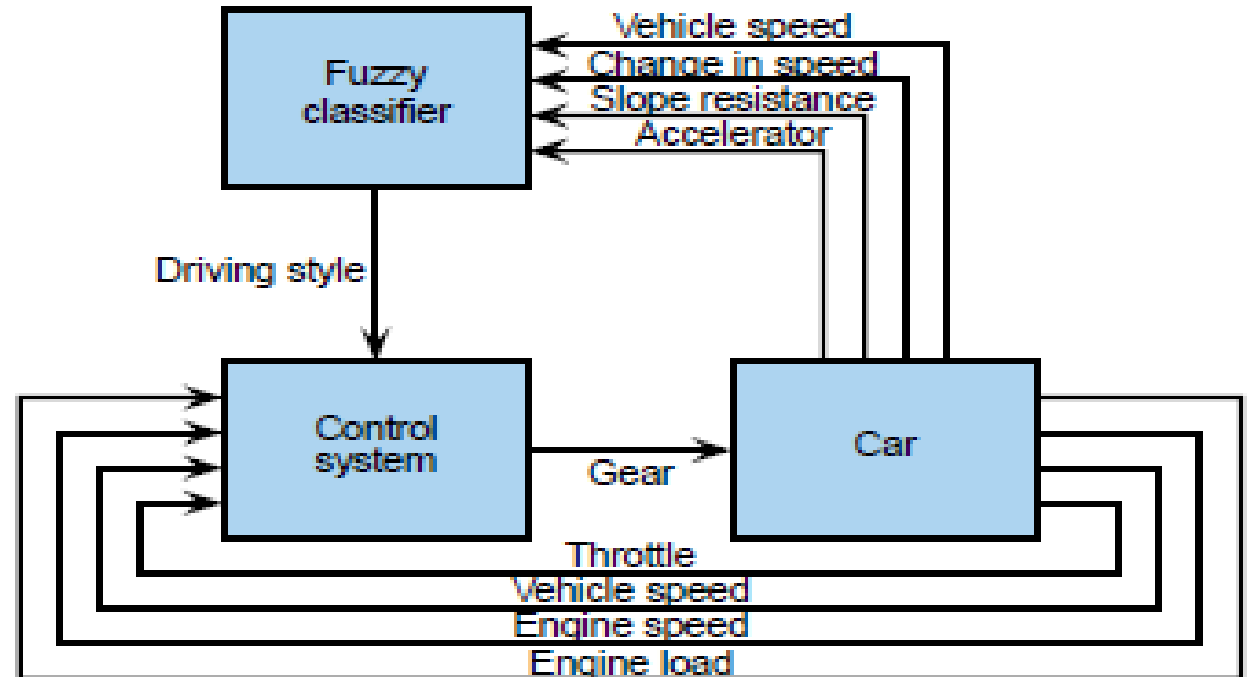
- Automatic gear selection behaviour
- Gear choice can be inferred from sensor readings
- Need to account for human factor



Human Decision Making

Volkswagen Direct-Shift Gearbox

- Two fuzzy systems are used:
 - Infer driving style
 - Select gear
- Gear selection based on:
 - Sensor data
 - Fuzzy judgement of current driving style



Human Decision Making

Volkswagen Direct-Shift Gearbox

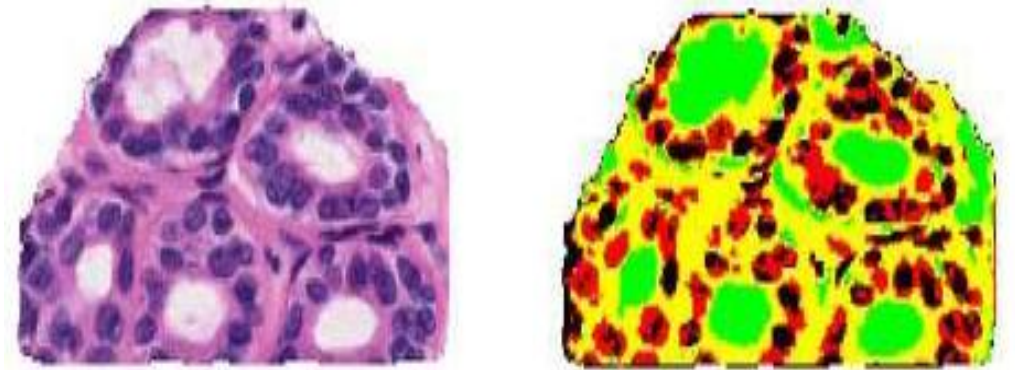
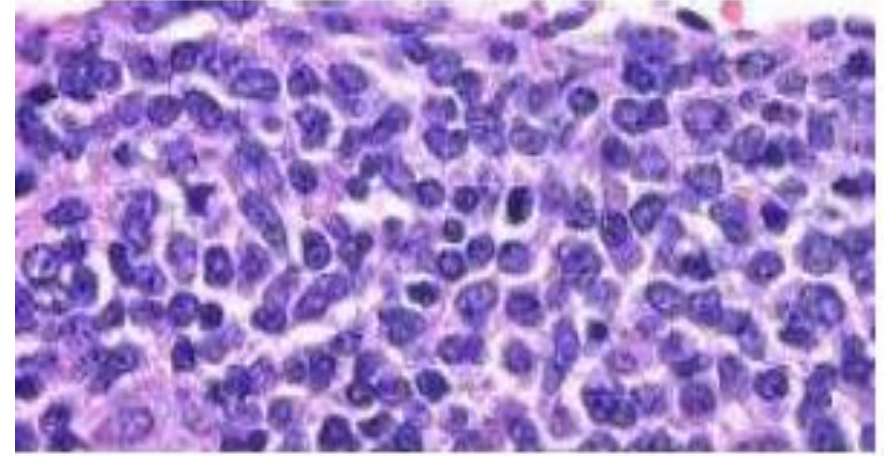
- Adaptive fuzzy systems
- Gradually adjusts the fuzzy sets
- Tailored to suit your personal driving style



Image Processing

Segmentation of Histopathology Images

- Identify regions of the image as:
 - Nuclei
 - Lumen
 - Cytoplasm
 - Classify tissue
- ❶ Set the number of classes n (3)
 - ❷ Initialise a fuzzy description of each
 - ❸ Find the set of fuzzy descriptions of n with the lowest overlap



Nuclei in red and black, lumen in green and cytoplasm in yellow

Application of Fuzzy Methods

- Useful wherever vagueness of uncertainty exists
- Relatively simple paradigm
- Not a panacea - good science is still the key
- Areas not mentioned:
 - White goods - fridges, freezers, washing machines
 - Camera anti-shake - Minolta and Canon
 - Scheduling - Seattle traffic light control system

Uncertainty and Vagueness

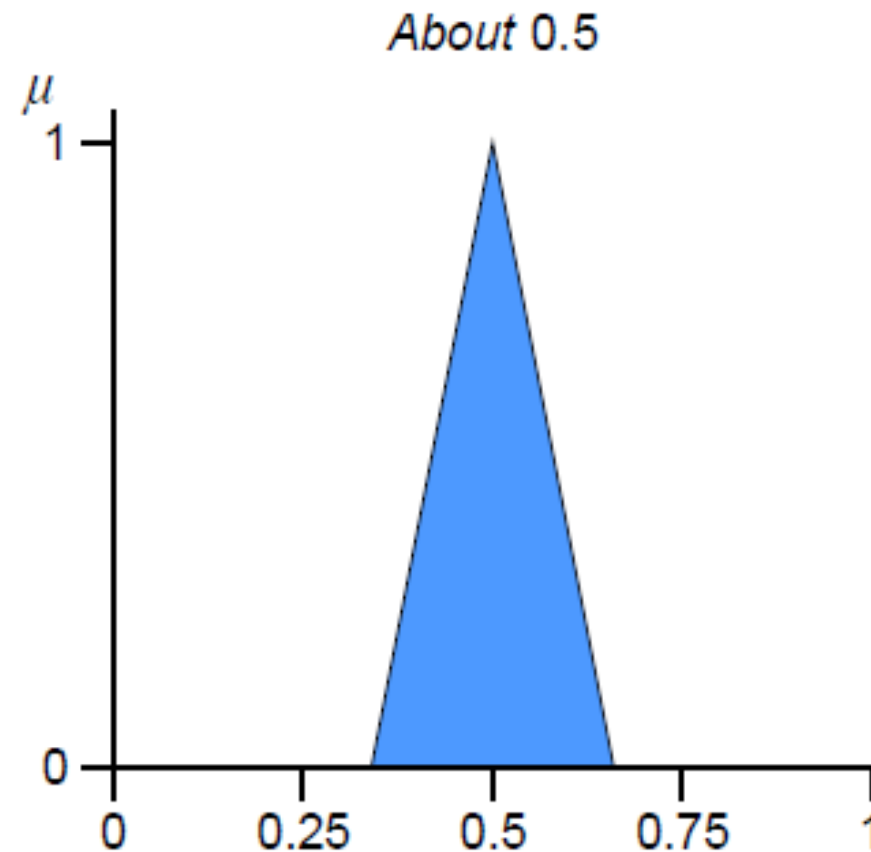
The Trouble with (Type-1) Fuzzy Sets

Fuzzy sets and systems:

- Vagueness
- Partial truth
- Degrees of set membership

But what about uncertainty?

- Alan is *0.5 Tall*
- 0.5 is crisp!
- Alan is *about 0.5 Tall*



The Trouble with (Type-1) Fuzzy Sets

Type-2 Fuzzy Sets:

- Set membership measured as a fuzzy number
- Alan is *about 0.5 Tall*
- Where *about 0.5* is a fuzzy set (number)
- DMU lead the world in this field
- Example type-2 fuzzy set - run program

A Personal View

- Uncertainly management is key
- Type-2 fuzzy systems have a **big role to play**
- Other extensions will also be important
- Computing with Words has potential
- Worth measured by applications

Summary

- Fuzzy sets are sets with soft boundaries
- Fuzzy logic performs inference on fuzzy sets
- Applied in a variety of areas
- Future developments are likely to be concerned with uncertainty models

Thank You....